

Talk2QSP: Deriving Executable Scenarios from Unstructured Literature via Human-in-the-Loop Agents

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Abstract

Quantitative Systems Pharmacology (QSP) models play an inherently interventional role in pharmaceutical research and development, functioning as executable causal systems for designing, evaluating, and replacing clinical trials. However, deploying QSP as an experimental planning engine remains constrained by the difficulty of translating unstructured literature descriptions of clinical or preclinical scenarios into reproducible, simulation-ready model interventions. Motivated by this issue, we propose an agent-based framework that operationalizes QSP models as intervention-ready experimental systems by automatically extracting and executing literature-derived scenarios. The framework combines semantic grounding of model entities with a large language model (LLM)-driven *Scenario Extractor* and a dual-agent *Scenario Mapper*. Rather than relying on opaque, single-shot reasoning, our pipeline converts free-text interventions into precise parameter configurations through discrete, verifiable work orders. Moreover, our dynamic Human-in-the-Loop (HITL) strategy empowers modelers to resolve biological ambiguities interactively. Across four diverse kinetic ordinary differential equation (ODE)/QSP models and seven Subject Matter Expert (SME)-curated literature scenarios, our model resolved all selected scenarios into correct executable parameter changes, including multi-dose interventions, unit conversions, no-op scenarios, and ambiguity-triggered

HITL cases, demonstrating that structured collaboration between experts and agentic systems can resolve scenarios that standalone Systems Biology Markup Language (SBML) reasoning LLM calls or prior agentic efforts operating on processed SBML data handle unreliably.

1. Introduction

QSP has become a central discipline for modeling and simulation efforts aimed at de-risking clinical development programs and assessing the translational potential of novel drug assets. QSP models integrate mechanistic modeling, pharmacokinetics/pharmacodynamics (PK/PD), and systems biology to quantitatively characterize drug–disease–biology interactions across multiple temporal and spatial scales. They are typically formulated as systems of dynamical equations encoding causal assumptions about molecular pathways, cellular interactions, and physiological processes, and are widely used in the pharmaceutical industry to inform target validation, dose and schedule optimization, biomarker selection, trial design, and translational inference from pre-clinical to clinical settings.

From a computational standpoint, QSP models are distinguished by large, high-dimensional, and uncertain parameter spaces that reflect biological variability, incomplete knowledge of system dynamics, and competing mechanistic hypotheses. As a result, meaningful deployment of QSP models relies on repeated execution across ensembles of parameterizations (such as virtual populations, hypothesis calibrations, or scenario sweeps) to produce distributions over outcomes rather than single point estimates. This execution pattern introduces challenges in scalability, reproducibility, and workflow orchestration that are largely independent of the underlying biology.

These challenges closely mirror core concerns in machine learning, including automated experiment management, sys-

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tematic exploration of complex parameter spaces, and standardized execution of computational graphs. This parallel motivates automated and scalable frameworks for managing, executing, and reusing QSP simulations across diverse parameter regimes.

Recent advances in large language model (LLM)-driven agentic systems are transforming biomedical research by shifting from task-specific automation toward integrated, multi-step analytical workflows. Agentic systems such as Talk2BioModels (Wehling et al., 2025), Talk2KnowledgeGraph (Singh et al., 2025), BiomniA1 (Huang et al., 2025), MEDEA (Sui et al., 2026), BioMedAgent (Bu et al., 2026), and The AI Scientist (Lu et al., 2026) demonstrate capabilities including mechanistic model interrogation, hypothesis generation, multi-step reasoning, omics-driven tool integration, and iterative execution to perform complex biomedical tasks. Collectively, these systems enable more scalable, interdisciplinary, and semi-autonomous computational pipelines, with the potential to accelerate scientific discovery (Li & Moore, 2026).

While these advances open new ways of building biomedical workflows, applying them to QSP requires confronting the specific characteristics of QSP models. QSP extends systems biology by integrating PK/PD (Knight-Schrijver et al., 2016), and its models are routinely exercised across many simulation scenarios, varying drug doses and dosing regimens to inform therapeutic decisions (Helmlinger et al., 2019). This breadth of scenarios is precisely what makes reproducibility both critical and difficult.

Ensuring reproducibility is critical for the reliability and validation of such models, as it allows independent verification of simulation outcomes (Tiwari et al., 2021). Reproducible models also support model reuse and extension of existing work, thereby accelerating scientific progress (Blinov et al., 2021). A key enabler of reproducibility is the ability to extract and share structured simulated scenarios, that is, the specific combination of species initialization, parameter settings, and dosing regimens used to reproduce figures or results of a publication. Without such structured representations, reproducing or building upon published models requires substantial manual effort, limiting the community’s ability to validate, compare, and extend models systematically.

While Systems Biology Markup Language (SBML) provides a standardized structure for model encoding (Hucka et al., 2015), it does not prescribe how simulation scenarios should be documented. In practice, extracting such scenarios from published models remains hindered by three key challenges. First, identifying relevant model components is often difficult, as information on initialized species or parameters for a given scenario is rarely provided in a structured, machine-readable format, and instead it must be inferred

from figures, tables, or free text in the accompanying publication. Second, interpreting model parameters requires considerable effort, since parameters are frequently under-annotated and lack clear biological definitions (Domínguez-Romero et al., 2024). Third, ensuring consistency of parameter values and units is non-trivial, as discrepancies between values and units in the SBML file and those reported in the publication require careful verification and conversion (Karr et al., 2022).

As a result, researchers must manually inspect reaction equations and model structures within SBML files to identify appropriate parameters and units, a process that is both time-consuming and error prone, where even minor inconsistencies can yield unreliable or misleading simulation results. Collectively, these challenges pose significant barriers to reproducibility, ultimately hindering the validation, interpretation, and reuse of QSP models (Tiwari et al., 2021; Blinov et al., 2021).

To bridge the gap between unstructured literature and computational models, we propose Talk2QSP, an agent-based framework for translating literature-derived scenarios into reproducible, simulation-ready QSP interventions. We make three primary contributions:

- **Entity Definer:** A semantic parsing component that translates complex QSP model parameters and biological species into human-readable, context-aware definitions.
- **Scenario Extractor:** An automated module that parses QSP literature to identify and extract the specific experimental scenarios evaluated within a given model.
- **Scenario Mapper:** An alignment framework that bridges the extracted text scenarios with the defined model entities, calculating the parameter adjustments required to computationally replicate published experimental results.

We design these three modules so that they can be deployed independently as standalone components or jointly as an end-to-end pipeline, offering a flexible and reproducible workflow for researchers in computational biology and bio-pharma.

2. Methodology

Our methodology is organized as a staged pipeline comprising three modules as illustrated in Figure 1. The Entity Definer first ingests a published SBML model and produces structured semantic artifacts (an *Entity Definitions Dictionary* and a compact *Minified Semantic Index*) that supply the biological grounding consumed by both downstream modules. The Scenario Extractor processes the associated

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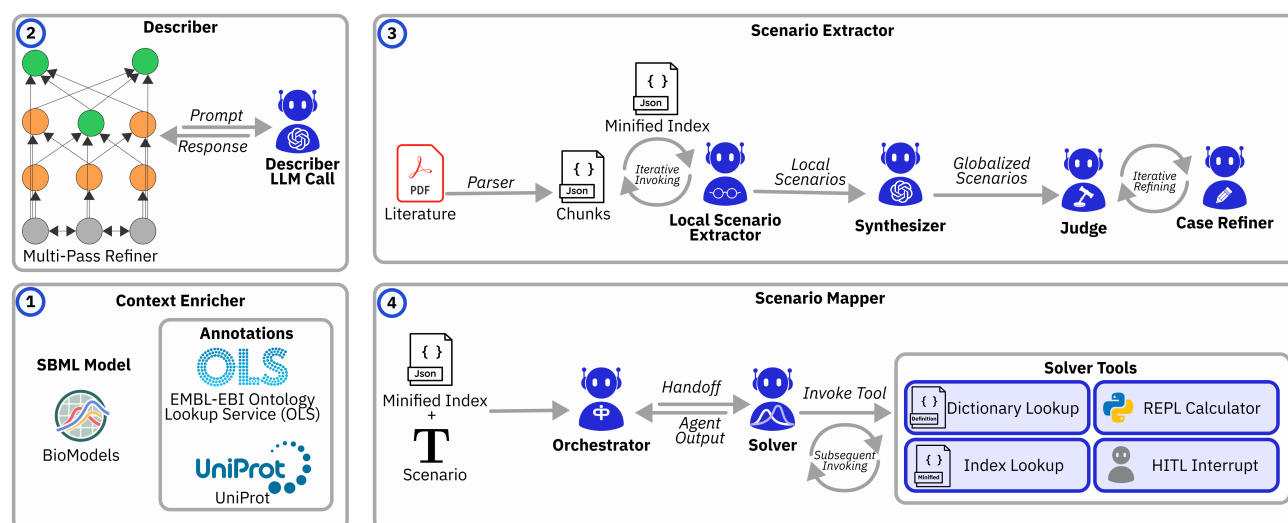


Figure 1. Overview of the proposed agentic framework. The pipeline comprises three modules: the Entity Definer, the Scenario Extractor, and the Scenario Mapper. The Entity Definer’s Context Enricher first aggregates reaction schemes, ontological annotations, and entity metadata from multiple sources, which the Descriptor uses to produce structured entity definitions; these artifacts are then consumed by both downstream modules. The Scenario Extractor parses the companion publication to identify all relevant QSP scenarios for the model of interest, and its structured output, together with the entity definitions, serves as input to the Scenario Mapper, which resolves the required parameter changes to produce simulation-ready configurations.

publication alongside the semantic index to identify and structure the experimental scenarios described in the paper. The Scenario Mapper then resolves each extracted scenario into simulation-ready parameter values by jointly leveraging both Entity Definer artifacts through an agentic workflow. Below, we detail each component in turn.

2.1. Entity Definer

The Entity Definer serves as the semantic bridge between raw mathematical specifications and natural language reasoning. This translation is essential not only for the downstream Scenario Extractor and Scenario Mapper, but also for assisting QSP modelers in interpreting complex, legacy model parameters. Taking a computational QSP SBML model as input (parsed via the BASICO (Bergmann, 2023) Python interface), the module systematically processes all internal variables. It executes a two-fold pipeline: first enriching the context of each variable, and subsequently processing it into a structured, human-readable JSON dictionary. This dictionary defines the biological identity, kinetic role, and compartment of every species and parameter.

Context Enricher: To prevent the language model from hallucinating biological intent from ambiguous mathematical syntax, we first employ a deterministic Context Enricher. This sub-module extracts raw mechanistic data, grouping all relevant reaction schemes and physical entity metadata (e.g., initial concentrations, units, and spatial com-

partments). Because reaction topology alone is frequently opaque (especially when variables rely on single-letter or non-self-explanatory names), we expand upon the annotation retrieval tool introduced by Talk2BioModels. Our system introduces robust, multi-library annotation support that actively queries external ontological databases such as UniProt (The UniProt Consortium, 2025) and the Ontology Lookup Service (OLS) (McLaughlin et al., 2025), ensuring that references to disparate databases are standardized and resolved via OLS. By fetching these verified annotations prior to inference, the enricher guarantees that the LLM is firmly grounded in established biological literature. The direct result of this stage is an intermediate *Enriched Mapping* artifact, which cleanly aggregates the governing reaction equations, passthrough initial values, and unified annotations to serve as the foundational context for the downstream synthesizer.

Descriptor and Multi-Pass Refinement: The core semantic synthesis is performed by the Descriptor, which transforms the Enriched Mapping into structured definitions. To prevent hallucinations, prompts enforce an authority hierarchy: external database annotations (e.g., UniProt, OLS) serve as ground truth for biological identity, overriding conflicting LLM inferences. Meanwhile, kinetic roles and units are derived directly from the governing equations and physical metadata.

Because biological networks are interconnected, definitions are produced through a multi-pass refinement loop with

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cumulative dependency tracking. In the first pass, every entity is processed concurrently via asynchronous LLM calls. Alongside its definition draft, the model must emit a self-evaluation tag (`COMPLETE` or `NEEDS_REFINEMENT`) and list any neighboring variables that lack biological grounding (e.g., referencing a bare symbol “L” without explaining what it represents).

Subsequent passes re-evaluate only the flagged entities. The Describer re-issues the original raw context, now augmented with a “Context from Related Entities” block. To keep the prompt compact, this block is restricted to the latest biological identity and definition of the missing dependencies, pulled from a running cache updated after every pass. Crucially, the entity’s own previous draft is never fed back. The LLM regenerates from scratch using the raw data and the newly enriched neighbor context, preventing earlier mistakes from being reinforced. The loop terminates when all entities reach `COMPLETE`, when no progress is observed, or when the `max_passes` threshold is reached.

The pipeline yields a suite of machine-actionable JSON artifacts tailored for different computational needs. The primary output is the Entity Definitions Dictionary, which profiles each variable and categorizes it as species or parameter. For each entity, it encapsulates the resolved biological identity, kinetic definition, physical compartments, units, and initial values, alongside execution metadata such as the required refinement passes and resolution status. The module also compiles a Minified Semantic Index. This compact artifact strips away verbose mathematical traces and execution metadata, flattening the ontology into a key-value index where each entity ID is mapped to a single, dense string summarizing its type, identity, and functional role. The Minified Index is designed to be injected into the limited context windows of the Scenario Extractor and Scenario Mapper, providing semantic grounding at low token cost.

2.2. Scenario Extractor

We design the Scenario Extractor to take as input a biomedical modeling paper in PDF format together with a lightweight context index containing short entity definitions and aliases used to stabilize terminology during prompting. Its output is a compact structured list of scenarios (`scenarios.json`), where each scenario is represented by a persistent identifier, a short natural-language description, and the text chunks that support it as evidence. At the current stage, we rely only on textual information; visual elements such as figures or diagrams are not explicitly used, although they could be incorporated in future multimodal extensions.

Before extraction, we convert the paper text into a deterministic, document-ordered sequence of chunks. Our chunking is rule-based and constrained by a maximum segment length

so that segmentation remains reproducible and each unit stays within a manageable context size for downstream inference. For extraction, consecutive chunks can be grouped into overlapping windows, which allows the model to process local context while preserving continuity across neighboring parts of the document.

ITERATIVE THREE-BLOCK DESIGN

We organize the extraction workflow into three high-level blocks that progressively move from local evidence detection to document-level consolidation. This design allows the system to first capture candidate scenarios with high sensitivity, then reconcile them globally, and finally perform a targeted quality-control pass when necessary.

Block 1: Local scenario extraction. Block 1 processes the paper sequentially through chunk windows and incrementally builds a set of candidate scenarios. At each step, the model receives the current local text span together with the scenario set accumulated so far, and proposes evidence-grounded updates such as adding a new candidate, revising an existing one, or discarding a weak duplicate.

The main role of this block is to maximize coverage while keeping each decision tied to a limited and explicit textual context. In practice, this stage acts as the recall-oriented component of the system: it aims to capture explicit study arms, regimens, intervention settings, simulation conditions, or comparison setups as they appear across the document.

Block 2: Paper-level synthesis. Block 2 performs a single paper-level synthesis pass over the candidates produced by Block 1 and the evidence associated with them. Its purpose is to move from a locally accumulated set of mentions to a coherent document-level representation of the scenarios described in the paper.

In this stage, the model consolidates semantically equivalent candidates, merges obvious duplicates, harmonizes wording, and reconstructs fragmented scenario descriptions only when the required elements are explicitly supported by the associated evidence chunks. Conceptually, this block acts as the global integration stage of the pipeline, preserving local evidence while producing a cleaner set of scenario-level outputs.

Block 3: Judge-guided refinement. Block 3 is a refinement loop designed as a lightweight judge-based framework. Rather than re-running full extraction or allowing unrestricted rewriting, the system reviews the synthesized scenario set globally and requests only localized, evidence-grounded edits when further changes are needed.

This stage separates two roles: a judge, which inspects the full set and identifies potential issues, and a refiner, which

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executes localized edits only when requested by the judge. If the scenario set is already satisfactory, the process stops. Otherwise, the judge sends specific refinement actions to the refiner, grounded on the scenarios under review and on their associated evidence chunks.

For example, the judge may determine that one scenario is too broad and ask the refiner to split it into two more specific scenarios, or it may indicate that two candidates should be merged because they describe the same intervention setting. The refiner then performs the requested operation using the chunk references and scenario identifiers provided in the refinement request, ensuring that edits remain localized, evidence-grounded, and traceable to the underlying text.

In the current implementation, this refinement loop is bounded to a maximum of three iterations. This final stage improves consistency, specificity, and overall coherence, while preserving the evidence-grounded structure built in the previous blocks.

2.3. Scenario Mapper

The Scenario Mapper is the final stage of our pipeline, bridging the gap between descriptive biology and mathematical simulation. It takes as input the free-text clinical pharmacology scenarios (whether generated by the Scenario Extractor or provided manually) alongside the Minified Semantic Index (generated by the Entity Definer). The output is a structured JSON array of model-ready parameter configurations, containing the exact numerical values, adjusted units, and computation histories required to replicate the literature scenario within the ordinary differential equation (ODE) Solver.

To handle the complex reasoning required for pharmacokinetic and pharmacodynamic translations, the Mapper leverages a dual-agent, orchestrator-solver architecture built on stateful graphs via LangGraph.

Orchestrator Agent: The pipeline is governed by the Orchestrator Agent, which acts as the semantic router. It analyzes the free-text scenario alongside the Minified Semantic Index, utilizing an LLM to identify the specific ODE parameters that require modification along with their relevant context. By injecting the Minified Index directly into the context window, this method avoids both the fragility of exact-string matching and the retrieval inaccuracies common in standard Retrieval-Augmented Generation (RAG) approaches, while keeping context sizes manageable. The Orchestrator formulates these modifications into discrete, structured "work orders". It feeds only one work order to the downstream solver at a time; this sequential dispatch improves user experience and prevents parallel execution conflicts during Human-in-the-Loop (HITL) interventions. This separation of concerns ensures that the Orchestrator focuses on *what*

needs to be computed, while the solver focuses on *how* to compute it.

Solver Agent Execution: Upon receiving a singular work order, the Solver Agent executes a multi-tool cognitive framework to resolve the final mathematical value. This process relies on iterative information retrieval, computation, and expert validation through the following integrated toolset:

- **Dictionary Lookup:** The agent establishes the target parameter's mathematical boundaries by executing an exact-syntax query against the Entity Definitions Dictionary. This retrieves the complete biological definition, initial value, and base units necessary to anchor the upcoming transformation.
- **Index Lookup:** To identify any missing auxiliary entities required to compute the target value, the agent semantically queries variables based on their biological purpose. A lightweight LLM, grounded by the Minified Semantic Index, processes these queries to return the exact variable names of the most relevant entities.
- **HITL Interrupt:** The agent may iterate between the Dictionary and Index lookups until all necessary inputs are gathered. If a critical requirement remains missing or a non-straightforward decision point is reached, the HITL mechanism is dynamically triggered, pausing execution to allow a domain expert to resolve the ambiguity.
- **Calculator (Python REPL):** Finally, the agent delegates the mathematical translation to a sandboxed Python REPL tool. By autonomously executing code to handle complex unit conversions (e.g., converting patient-weight-adjusted mass into molar concentrations), the Calculator produces deterministic outputs, avoiding the arithmetic errors common in LLM outputs.

Together, this integrated toolset ensures the Solver Agent can autonomously bridge the semantic and mathematical gaps required to execute literature-derived interventions. Upon resolving a parameter, the Solver triggers a terminal formatting node that structures its execution history into a validated JSON object containing the exact numerical values and the explanatory biological context. These individual outputs are then returned to the Orchestrator, which aggregates all completed work orders into a consolidated parameter update summary, presenting the final simulation-ready state back to the user.

3. End-to-End Evaluation

Implementation Details: The Entity Definer's Describer stage, the Scenario Extractor (Blocks 1–3), and the Sce-

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nario Mapper’s Orchestrator and Solver agents all use OpenAI GPT-4.1-mini with `temperature=0` and `seed=42`, to maximize reproducibility within the limits of the OpenAI API. Within the Solver, the simpler task of semantic search against the Minified Index is served by a smaller model (GPT-4.1-nano) through the lightweight Index Lookup tool. The Describer is configured with up to eight refinement passes and up to ten concurrent LLM calls per pass for processing the interconnected reactions produced by the Stage-1 enriched mapping. The Scenario Extractor processes the chunked paper text (approximately 320 tokens per chunk) with a window size of two and an overlap of one, with Block 3 enabled for up to three judge-refiner rounds. The Orchestrator of the Scenario Mapper dispatches work orders sequentially to support HITL clarification.

Evaluation Framework: While intrinsic evaluation isolates the performance of individual components, evaluating an integrated agentic workflow requires an extrinsic, end-to-end approach to measure system utility and account for compounding variables. To demonstrate pipeline viability, we conducted end-to-end case studies translating raw computational models and their accompanying literature into executable parameter states.

Pipeline Execution: The evaluation followed a sequential execution mirroring real-world deployment:

1. **Semantic Grounding:** The target model was processed by the Entity Definer to generate the required semantic artifacts.
2. **Scenario Extraction:** The accompanying publication was processed by the Scenario Extractor to surface candidate experimental scenarios based on the text.
3. **HITL Filtering:** A domain expert reviewed the extracted scenarios, mirroring the intended user workflow. Scenarios that were out-of-scope or non-actionable were discarded, and only relevant scenarios were selected for downstream execution.
4. **Parameter Mapping:** The selected free-text scenarios were fed into the Scenario Mapper to calculate the modified parameter configurations.

Ground Truth Mapping: To quantify the accuracy of the end-to-end pipeline, the outputs of the Scenario Mapper were evaluated against the established ground truth for each scenario (derived from the original authors’ simulation settings). We evaluated the pipeline based on the resulting “Delta”, reporting whether the system successfully identified the correct target parameters, executed necessary unit conversions, and mapped them to the exact numerical values required to replicate the literature scenario.

Results: We evaluated the workflow on four QSP models spanning a deliberately diverse set of scenarios: multi-dose interventions (BioModels 512, 788), multi-parameter interventions requiring intermediate computations (BioModels 537, 788), and scenarios that require *no* parameter change but whose phrasing could plausibly mislead the mapper (BioModels 537 and 913). For each model, the Entity Definer first produced its semantic artifacts as described in the Methodology, the Scenario Extractor returned the full set of candidate scenarios, a Subject Matter Expert (SME) selected the relevant ones for downstream execution, and the Scenario Mapper resolved each selected scenario into parameter configurations. The mapped outputs were then compared against an SME-curated ground truth that reproduces the published figures and results. As reported in Table 1, the pipeline passed all seven end-to-end tests, with at most one HITL clarification per parameter, demonstrating that the Mapper consistently aligns extracted scenarios with the correct entities and target values.

BioModels 512 (Osteoarthritic Pain) exercises the Mapper’s multi-dose handling: the Orchestrator emits a single work order for the dosing parameter and the Solver resolves the seven dose levels in sequence. The HITL interrupt is triggered once, exclusively to confirm unit is `mg`, since the SBML model does not declare a unit for the dose entity and the surrounding SBML context alone is insufficient to disambiguate it.

BioModels 537 (IBD (Crohn’s disease)) stresses multi-step pharmacological reasoning. In the first two scenarios, the Solver correctly converts the body-weight-normalized dose (8 mg/kg) using an average human body weight of 70 kg, and computes `kIL6RUnbind` from the relation

$$k_{\text{off}} = K_d \cdot k_{\text{on}},$$

by retrieving the missing association rate constant (k_{on}) from the model and combining it with the literature-reported K_d . This illustrates the Solver’s ability to chain Dictionary lookups, Index lookups, and the Calculator tool to resolve parameters that are coupled through reaction kinetics. The correct selection of `Dose` for the every-4-week regimen and `DoseQ2W` for the every-2-week regimen further demonstrates the Orchestrator’s ability to disambiguate among similarly-named parameters by attending to specific contextual cues (here, the dosing frequency) in the scenario text. The remaining two BioModels 537 rows verify that the system also handles scenarios that require *no* parameter change. In the placebo scenario, the `Dose` parameter is correctly set to 0. In the “varying dose” scenario, the Orchestrator does identify `Dose` as a candidate, but, because the SBML model does not declare a dose unit and various dose is ambiguous, the Solver triggers a single HITL clarification to confirm the change and implicit `mg` assumption before returning the parameter’s value.

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Table 1. End-to-end evaluation results of the full pipeline across four QSP models and seven scenarios. Each row corresponds to a scenario extracted from the model’s companion publication. *Extracted Scenario Text* is the free-text scenario as identified by the Scenario Extractor. *Expected Entity Change* lists the model parameters and target values the Scenario Mapper must resolve. *o3 SBML Baseline* reports the parameter mapping produced by a single-shot OpenAI o3 call given only the raw SBML context; cells marked *Pass* indicate the baseline matches the expected entity change, otherwise the parameters and values returned by the baseline are listed. *Talk2Biomodels Baseline* reports the parameter mapping produced by the Talk2Biomodels agentic system on the same scenarios, evaluated analogously. *Talk2QSP* indicates whether the full pipeline resolved the scenario autonomously (*Pass*) or required HITL clarification (*HITL Pass*).

BioModel ID	Extracted Scenario Text	Expected Entity Change	o3 SBML Baseline	Talk2Biomodels Baseline	Talk2QSP Results
512	Model calibration simulation of percentage cannabinoid receptor (CB1) occupancy in the brain at doses of 0.1 mg, 0.3 mg, 1 mg, 3 mg, 10 mg, 20 mg, and 40 mg, as shown in Figure 5.	Dose: 0.1; 0.3; 1; 3; 10; 20; 40	Pass	Pass	Pass
537	Treatment with 8 mg/kg anti-IL-6R α antibody administered intravenously every 4 weeks with Kd of 1 nmol/l.	Dose: 560; kIL6RUnbind: 1000	Pass	Dose: 8; kIL6RBind: 1; Ab: 8	Pass
	Treatment with 8 mg/kg anti-IL-6R α antibody administered intravenously every 2 weeks with Kd of 1 nmol/l.	DoseQ2W: 560; kIL6RUnbind: 1000	Dose: 560; parameter_1: 560; mw1...01ead: 1000	Dose: 8; DoseQ2W: 8; kIL6RBind: 1; Ab: 8; Ab.sR: 8; Ab.sR_IL6: 8	Pass
	Placebo treatment;	Dose: 0	Pass	Dose: 0; CRP: 0; sR_IL6: 0; IL6: 0	Pass
	Dose-response experiment under every 4-week intravenous dosing for 12 weeks with anti-IL-6 and anti-IL-6R α antibodies targeting both transmembrane and soluble IL-6R α , simulating effects on serum CRP, pSTAT3 activity in the GI tract, and IL-6 levels in the GI tract and serum, varying dose and binding affinities.	No parameter change required	Dose: 10; 30; 100; 300; 600; kIL6RUnbind: 25; 2.5; 0.25	Dose: 600; kIL6RBind: 1500; kdegIL6: 20; IL6: 0.0001; CRP: 100	HITL Pass
788	Simulations with the full BsAb model for 50 mg and 250 mg intravenous doses without peripheral compartment, evaluating area under the curve (AUC) of BsAb concentration and ternary complex (TC) formation over time (Figure 3).	k_12: 0; k_21: 0; C_free: 50 mg = 335 nM; 250 mg = 1675 nM	k_12: 0; k_21: 0; C_free: 16.67; 83.33	C_free: 250; R_A: 0	HITL Pass
913	Quadratic control simulation of IL-2 treatment (VI) and chemotherapy treatment (VM) with detectable initial tumor conditions ($T = 10^7$), tracing tumor, natural killers, CD8 $^+$ T cells, IL-2, and circulating lymphocytes over 5 days, as shown in Figure 1.	No parameter change required	V_I: 20000; V_M: 1	N: 10^6 ; C: 3×10^{10} ; I: 5×10^3 ; L: 4×10^3 ; g_I : 3×10^7 ; g_T : 1.5×10^3 ; α_1 : 2×10^4 ; β : 0.008	HITL Pass

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Table 2. Summary of QSP models used for benchmark. Each entry lists the BioModels identifier, the associated disease and the corresponding reference

BioModels ID	Disease	Reference
BIOMD0000000512	Osteoarthritic Pain	(Benson et al., 2014)
BIOMD0000000537	IBD (Crohn’s disease)	(Dwivedi et al., 2014)
BIOMD0000000788	Bispecific antibodies (BsAbs) therapies	(Schropp et al., 2019)
BIOMD0000000913	Generic solid tumors (tumor immune interactions)	(de Pillis et al., 2008)

BioModels 913 (Generic solid tumors (tumor immune interactions)) highlights a subtler failure mode that the HITL design is meant to catch: the SBML unit for the extracted parameter (T, in mmol/ml) disagrees with the quantity reported in the publication, which is a tumor cell population (i.e., a count, not a concentration). The Solver pauses to double-check the conversion, giving the SME the ability to override the proposed decision. This case also exemplifies a broader modeling practice (where default SBML units are retained as placeholders for raw numerical values rather than performing strict unit conversions to reflect the paper’s narrative), which we discuss as a direction for future work.

BioModels 788 (Bispecific antibodies (BsAbs) therapies) further demonstrates the pipeline’s robustness in handling these unit disparities and the aforementioned practice of retaining default SBML units as placeholders. First, to capture the “without peripheral compartment” condition, the Mapper correctly identifies that the transition rate constants k_{12} and k_{21} must be set to 0. The scenario also requires translating intravenous doses (50 mg and 250 mg) into the model’s free-concentration entity, C_{free} , which the SBML model strictly defines in nmol/L (nM). The Solver correctly recognizes that this gap between mass and molar concentration cannot be closed from the SBML context alone and proposes the structural conversion:

$$C_{free} = \frac{\text{Dose} \times 10^6}{MW \cdot V_c},$$

triggering a HITL interrupt to request the missing molecular weight (MW) and central compartment volume (V_c) from the SME. If the SME supplies the conversion factor explicitly stated in the paper’s methodology (1 mg = 6.7 nM), the Calculator deterministically computes the mathematically accurate concentrations (335 nM and 1675 nM). Alternatively, replicating the publication’s graphical results and published parameter information reveals that the original authors used the default SBML units as placeholders, directly inputting the raw mass values (50 and 250) into their simulation software. When the SME instructs the agent to bypass the mathematical conversion to mirror this placeholder practice, the Mapper successfully passes the raw values directly to the parameter configuration. This case

demonstrates the system’s flexibility in accommodating both mathematically rigorous derivations and strictly empirical, literature-matched replications based on dynamic human guidance.

Reasoning-LLM baseline. To probe whether a strong off-the-shelf reasoning LLM can substitute for the structured agentic workflow, we benchmarked OpenAI o3, a frontier reasoning model with strong multi-step reasoning and tool-use capabilities, on the same seven scenarios, providing only the raw SBML XML and the scenario text in a single call (no Entity Definer artifacts, no Orchestrator/Solver decomposition, no HITL hooks; we deliberately did not pin sampling parameters in order to characterize run-to-run variability). The *o3 SBML Baseline* column of Table 1 reports a representative run, and we repeated the calls three times per scenario. Only BioModels 512 (dose titration) and the BioModels 537 placebo are recovered consistently (3/3). The remaining scenarios exhibit run-to-run instabilities that mirror our design motivations: fabrication of parameter names that do not exist in the SBML model on BioModels 537 (e.g. `ModelValue_98`, `mw1c4bc9c3...`); missed body-weight conversions on the every-2-week scenario, where `DoseQ2W` is resolved to 8 mg/kg in one run and 560 mg in another; silent omission of `C_free` on BioModels 788, which appears in only one of three runs and with an incorrect mass-to-concentration conversion; and, on BioModels 913, fabrication of non-existent parameters `V_I` and `V_M` (neither of which appears in the SBML), even though the scenario requires no parameter change at all, with `V_I` additionally varying from 4 000 to 50 000 000 across runs (a four-orders-of-magnitude swing). These failure modes correspond directly to our three contributions: grounded, validated entity naming (Entity Definer), discrete and verifiable work orders (Orchestrator), and ambiguity-triggered HITL interrupts (Solver). At the same time, the baseline confirms that o3 itself is a competent unit reasoner that we view as a natural drop-in primitive within our agentic framework wherever stronger reasoning is required, rather than as a replacement for the agentic decomposition.

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Agentic-tool-use baseline (Talk2Biomodels). We further benchmarked Talk2Biomodels on the same seven scenarios with GPT-4.1-mini as the underlying LLM, asking it which parameters and species to change and to what values. Because Talk2Biomodels enumerates entities directly from the uploaded SBML through `basico`, the baseline does not fabricate non-existent parameter names (the failure mode observed for o3) and recovers the BioModels 512 dose titration. Across the remaining six scenarios we observe four recurring failure modes, all reducing to the absence of an explicit reasoning and HITL layer:

Missed unit conversions: “8 mg/kg” written as 8 into the absolute Dose; “250 mg” written as 250 into the nmol/L species `C_free` (incidentally matching the publication’s mg-as-placeholder convention, but reached without unit reasoning).

Kinetic-constant conflation: equilibrium constant K_d used in place of association rate k_{on} .

Hardcoded endogenous state: CRP, IL6, `sR_IL6` zeroed in the placebo arm; the Q2W dose pasted into Dose, DoseQ2W, and the antibody complexes `Ab_sR/Ab_sR_IL6` (breaking mass balance at $t = 0$); eight quantities altered on BioModels 913 where no change is required.

Collapsed parameter sweeps: no 50/250 mg arm on BioModels 788, with `k_12/k_21` left active despite the “without peripheral compartment” instruction.

Grounded entity access through `basico` is therefore necessary but not sufficient: the remaining gap is precisely what our Entity Definer, Orchestrator/Solver work orders, and HITL interrupts are designed to close.

4. Discussion and Future Work

In this work, we introduce an agentic framework that reduces the friction of reusing and validating published QSP models. By providing comprehensive semantic grounding of SBML files, automatically extracting experimental scenarios from accompanying literature, and dynamically mapping free-text descriptions to executable parameter states, our pipeline transforms static publications into interactive, simulation-ready systems. This capability addresses a critical bottleneck in computational biology, enabling researchers to seamlessly replicate and iterate upon existing models using natural language. Across both the o3 single-shot reasoning baseline and the Talk2Biomodels agentic baseline, the seven scenarios are recovered only partially, indicating that the combination of grounded entity access, discrete work orders, and HITL interrupts in our pipeline materially closes the gap.

While our current pipeline establishes a robust foundation for automated model management, several avenues remain

for expanding its capabilities. Currently, the Scenario Extractor and Mapper rely exclusively on textual data. However, critical experimental conditions are frequently embedded within complex visual modalities (such as charts and figures) without explicit textual descriptions. Future iterations will integrate multimodal vision-language capabilities to process figure captions and graphical data directly, leveraging open-source vision-language models such as Pixtral (Agrawal et al., 2024) and Qwen2.5-VL (Bai et al., 2025). Furthermore, the Entity Definer currently grounds semantic meaning in each entity’s intrinsic attributes (name, type, unit, and initial value), its reaction context (kinetic schemes, mappings, and rate-law functions), and MIRIAM ontology annotations retrieved from UniProt and the OLS. Future iterations will broaden this coverage to include complementary intra-model signals such as SBML events, assignment rules, and parameter-level notes (useful, for instance, for dosing-control switches in event-driven models like BioModels 620), as well as document-level evidence for cases where essential parameter context resides only in the publication narrative (e.g., BioModels 913).

Additionally, the present architecture of the Scenario Mapper operates on a one-to-one mapping paradigm, which efficiently handles discrete interventions. Translating aggregated or multidimensional scenarios, such as complex parameter sweeps or changing interconnected groups of entities, will be a key functional improvement for the solver agent.

Finally, the broader transition toward agentic biomedical systems requires standardized evaluation frameworks. Because no large-scale, annotated dataset of QSP scenarios currently exists, our end-to-end evaluation was necessarily constrained to a manually curated set of seven scenarios designed to demonstrate the viability of the architecture. Developing a comprehensive benchmark dataset (consisting of diverse QSP models, their paired publications, and validated sets of extracted scenarios with exact parameter modifications) is an essential next step. Establishing such a benchmark will not only provide a rigorous grounding for evaluating and refining our modules at scale, but also serve as a vital open resource for the wider community developing generative AI tools for systems biology.

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Conflict of Interests

AK, JP, AS, PP, and TA are Sanofi employees and may hold shares and/or stock options in the company.

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